Year 8

REVISION PACK

Unit 3

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Unit 3 · Simplifying and expanding

Q	Working	Answer
collecting like terms and expanding brackets		
1.1	**(a)** $2y^2 + y^2 = 3y^2$ **(b)** $15j + 20 - 4j - 6$ $= 11j + 14$ **(c)** $5x + 10y + 3y - 3x$ $= 2x + 13y$	**(a)** $3y^2 + 3y + 6$ **(b)** $11j + 14$ **(c)** $2x + 13y$
1.2	**(a)** $5a^2 + 3a^2 = 8a^2$ **(b)** $12j + 6 - 2j - 3$ $= 10j + 3$ **(c)** $3x + 15y + 2y - 2x$ $= x + 17y$	**(a)** $8a^2 + 2a + 1$ **(b)** $10j + 3$ **(c)** $x + 17y$
1.3	**(a)** $4m^2 + 2m^2 = 6m^2$ **(b)** $12j + 6 - 6j - 9$ $= 6j - 3$ **(c)** $5x + 15y + 4y - 4x$ $= x + 19y$	**(a)** $6m^2 + 7m + 9$ **(b)** $6j - 3$ **(c)** $x + 19y$
1.4	**(a)** $p^2 + 6p^2 = 7p^2$ **(b)** $10j + 8 - 8j - 12$ $= 2j - 4$ **(c)** $6x + 12y + 2y - 2x$ $= 4x + 14y$	**(a)** $7p^2 + 4p + 5$ **(b)** $2j - 4$ **(c)** $4x + 14y$

1.5	**(a)** $3t^2 + 4t^2 = 7t^2$ **(b)** $15j + 10 - 4j - 6$ $= 11j + 4$ **(c)** $2x + 14y + 4y - 4x$ $= -2x + 18y$	**(a)** $7t^2 + 6t + 2$ **(b)** $11j + 4$ **(c)** $-2x + 18y$
1.6	**(a)** $7n^2 + 2n^2 = 9n^2$ **(b)** $8j + 24 - 6j - 9$ $= 2j + 15$ **(c)** $7x + 21y + 5y - 5x$ $= 2x + 26y$	**(a)** $9n^2 + n + 8$ **(b)** $2j + 15$ **(c)** $2x + 26y$

Unit 3 · Using a formula

Q	Working	Answer
substituting and solving		
2.1	**(a)** $0^2 = 20^2 + 2 \times (-4) \times s$ $0 = 400 - 8s$ $8s = 400$ $s = 50$ **(b)** $10^2 = 6^2 + 2 \times 2 \times s$ $100 = 36 + 4s$ $4s = 64$ $s = 16$	**(a)** $s = 50$ **(b)** $s = 16$
2.2	**(a)** $0^2 = 12^2 + 2 \times (-3) \times s$ $0 = 144 - 6s$ $6s = 144$ $s = 24$ **(b)** $13^2 = 5^2 + 2 \times 4 \times s$ $169 = 25 + 8s$ $8s = 144$ $s = 18$	**(a)** $s = 24$ **(b)** $s = 18$

2.3	**(a)** $0^2 = 14^2 + 2 \times (-7) \times s$ $0 = 196 - 14s$ $14s = 196$ $s = 14$ **(b)** $9^2 = 3^2 + 2 \times 6 \times s$ $81 = 9 + 12s$ $12s = 72$ $s = 6$	**(a)** $s = 14$ **(b)** $s = 6$
2.4	**(a)** $0^2 = 30^2 + 2 \times (-10) \times s$ $0 = 900 - 20s$ $20s = 900$ $s = 45$ **(b)** $8^2 = 4^2 + 2 \times 3 \times s$ $64 = 16 + 6s$ $6s = 48$ $s = 8$	**(a)** $s = 45$ **(b)** $s = 8$
2.5	**(a)** $0^2 = 16^2 + 2 \times (-8) \times s$ $0 = 256 - 16s$ $16s = 256$ $s = 16$ **(b)** $12^2 = 2^2 + 2 \times 5 \times s$ $144 = 4 + 10s$ $10s = 140$ $s = 14$	**(a)** $s = 16$ **(b)** $s = 14$
2.6	**(a)** $0^2 = 9^2 + 2 \times (-2) \times s$ $0 = 81 - 4s$ $4s = 81$ $s = 20.25$ **(b)** $11^2 = 1^2 + 2 \times 10 \times s$ $121 = 1 + 20s$ $20s = 120$ $s = 6$	**(a)** $s = 20.25$ **(b)** $s = 6$

Unit 3 · Factorising

Q	Working	Answer
factorising fully		
3.1	**(a)** $\text{HCF} = 5$ **(b)** $\text{HCF} = d$ **(c)** $\text{HCF} = x$ **(d)** $\text{HCF} = 5x^2$ **(e)**	**(a)** $5(4x + 7)$ **(b)** $d(3d - 7)$ **(c)** $x(12 + y)$ **(d)** $5x^2(2 + 3x)$ **(e) No: the HCF is 10, so 10(3 - 2t)**
3.2	**(a)** $\text{HCF} = 7$ **(b)** $\text{HCF} = k$ **(c)** $\text{HCF} = x$ **(d)** $\text{HCF} = 3x^2$ **(e)**	**(a)** $7(2y + 3)$ **(b)** $k(5k - 3)$ **(c)** $x(7 + y)$ **(d)** $3x^2(2 + 3x)$ **(e) No: the HCF is 6, so 6(3 + 4a)**
3.3	**(a)** $\text{HCF} = 6$ **(b)** $\text{HCF} = m$ **(c)** $\text{HCF} = x$ **(d)** $\text{HCF} = 4x^2$ **(e)**	**(a)** $6(4a + 5)$ **(b)** $m(4m - 9)$ **(c)** $x(11 + y)$ **(d)** $4x^2(2 + 5x)$ **(e) No: the HCF is 10, so 10(2 - 3n)**
3.4	**(a)** $\text{HCF} = 5$ **(b)** $\text{HCF} = h$ **(c)** $\text{HCF} = x$ **(d)** $\text{HCF} = 5x^2$ **(e)**	**(a)** $5(2p + 5)$ **(b)** $h(3h - 8)$ **(c)** $x(5 + y)$ **(d)** $5x^2(3 + 2x)$ **(e) No: the HCF is 8, so 8(2x + 3)**
3.5	**(a)** $\text{HCF} = 9$ **(b)** $\text{HCF} = t$ **(c)** $\text{HCF} = x$ **(d)** $\text{HCF} = 7x^2$ **(e)**	**(a)** $9(2b + 3)$ **(b)** $t(6t - 5)$ **(c)** $x(20 + y)$ **(d)** $7x^2(2 + 3x)$ **(e) No: the HCF is 12, so 12(3 - 4y)**

3.6	**(a)** $\text{HCF} = 8$ **(b)** $\text{HCF} = w$ **(c)** $\text{HCF} = x$ **(d)** $\text{HCF} = 3x^2$ **(e)**	**(a)** $8(2c + 5)$ **(b)** $w(7w - 2)$ **(c)** $x(13 + y)$ **(d)** $3x^2(3 + 2x)$ **(e) No: the HCF is 14, so 14(2 + 3b)**

Unit 3 · Indices in algebra

Q	Working	Answer
simplifying with the laws of indices		
4.1	**(a)** $6 + 3 = 9$ **(b)** $12 - 5 = 7$ **(c)** $4 \times 2 = 8$ $x^3 \times x = x^4$ **(d)** $3^3 = 27$ $b^{2\times3} = b^6$	**(a)** n^9 **(b)** c^7 **(c)** $8x^4y$ **(d)** $27a^3b^6$
4.2	**(a)** $2 + 7 = 9$ **(b)** $10 - 4 = 6$ **(c)** $5 \times 3 = 15$ $x^2 \times x = x^3$ **(d)** $4^2 = 16$ $b^{3\times2} = b^6$	**(a)** p^9 **(b)** k^6 **(c)** $15x^3y$ **(d)** $16a^2b^6$
4.3	**(a)** $4 + 4 = 8$ **(b)** $8 - 3 = 5$ **(c)** $6 \times 2 = 12$ $x^4 \times x = x^5$ **(d)** $2^3 = 8$ $b^{5\times3} = b^{15}$	**(a)** t^8 **(b)** y^5 **(c)** $12x^5y$ **(d)** $8a^3b^{15}$

4.4	**(a)** $9 + 2 = 11$ **(b)** $11 - 6 = 5$ **(c)** $2 \times 7 = 14$ $x^5 \times x = x^6$ **(d)** $5^2 = 25$ $b^{2\times 2} = b^4$	**(a)** h^{11} **(b)** d^5 **(c)** $14x^6y$ **(d)** $25a^2b^4$
4.5	**(a)** $3 + 8 = 11$ **(b)** $15 - 9 = 6$ **(c)** $3 \times 4 = 12$ $x^2 \times x = x^3$ **(d)** $3^2 = 9$ $b^{4\times 2} = b^8$	**(a)** w^{11} **(b)** g^6 **(c)** $12x^3y$ **(d)** $9a^2b^8$
4.6	**(a)** $5 + 5 = 10$ **(b)** $7 - 2 = 5$ **(c)** $7 \times 2 = 14$ $x^3 \times x = x^4$ **(d)** $10^3 = 1000$ $b^{2\times 3} = b^6$	**(a)** r^{10} **(b)** z^5 **(c)** $14x^4y$ **(d)** $1000a^3b^6$

Unit 3 · Expressions and shapes

Q	Working	Answer
perimeter and area of a rectangle		
5.1	**(a)** $2(3d + 2) + 2(d + 5) = 30$ $8d + 14 = 30$ $8d = 16$ $d = 2$ **(b)** $\text{length} = 8,\ \text{width} = 7$ $8 \times 7 = 56$	**(a)** $d = 2$ **(b) 56 cm²**

5.2	**(a)** $2(2d+1)+2(d+3)=32$ $6d+8=32$ $6d=24$ $d=4$ **(b)** $\text{length}=9,\ \text{width}=7$ $9\times 7=63$	**(a)** $d=4$ **(b) 63 cm²**
5.3	**(a)** $2(4d+1)+2(d+2)=36$ $10d+6=36$ $10d=30$ $d=3$ **(b)** $\text{length}=13,\ \text{width}=5$ $13\times 5=65$	**(a)** $d=3$ **(b) 65 cm²**
5.4	**(a)** $2(3d+4)+2(2d+1)=50$ $10d+10=50$ $10d=40$ $d=4$ **(b)** $\text{length}=16,\ \text{width}=9$ $16\times 9=144$	**(a)** $d=4$ **(b) 144 cm²**
5.5	**(a)** $2(5d+2)+2(d+6)=52$ $12d+16=52$ $12d=36$ $d=3$ **(b)** $\text{length}=17,\ \text{width}=9$ $17\times 9=153$	**(a)** $d=3$ **(b) 153 cm²**
5.6	**(a)** $2(2d+5)+2(d+1)=30$ $6d+12=30$ $6d=18$ $d=3$ **(b)** $\text{length}=11,\ \text{width}=4$ $11\times 4=44$	**(a)** $d=3$ **(b) 44 cm²**

Unit 3 · Identities

Q	Working	Answer
showing two expressions are the same		
6.1	**(a)** $x^2 \times x + x^2 \times 2x^2 + 4x^3$ $= x^3 + 2x^4 + 4x^3$ $= 5x^3 + 2x^4$ **(b)** $4x + 12 + 2x - 6$ $= 6x + 6$	**(a) shown** **(b) shown**
6.2	**(a)** $x^2 \times x + x^2 \times 4x^2 + 2x^3$ $= x^3 + 4x^4 + 2x^3$ $= 3x^3 + 4x^4$ **(b)** $5x + 5 + 3x - 3$ $= 8x + 2$	**(a) shown** **(b) shown**
6.3	**(a)** $x^2 \times x + x^2 \times 5x^2 + 3x^3$ $= x^3 + 5x^4 + 3x^3$ $= 4x^3 + 5x^4$ **(b)** $2x + 8 + 3x - 12$ $= 5x - 4$	**(a) shown** **(b) shown**
6.4	**(a)** $x^2 \times x + x^2 \times 3x^2 + 6x^3$ $= x^3 + 3x^4 + 6x^3$ $= 7x^3 + 3x^4$ **(b)** $6x + 12 + 4x - 8$ $= 10x + 4$	**(a) shown** **(b) shown**
6.5	**(a)** $x^2 \times x + x^2 \times 6x^2 + 2x^3$ $= x^3 + 6x^4 + 2x^3$ $= 3x^3 + 6x^4$ **(b)** $3x + 15 + 2x - 10$ $= 5x + 5$	**(a) shown** **(b) shown**

6.6	**(a)** $x^2 \times x + x^2 \times 7x^2 + 4x^3$ $= x^3 + 7x^4 + 4x^3$ $= 5x^3 + 7x^4$ **(b)** $7x + 7 + 5x - 5$ $= 12x + 2$	**(a) shown** **(b) shown**